

Groq를 이용한 Chatbot 예제

Groq

- Groq는 AI 모델을 빠르고 저렴하게 실행(Inference)하기 위해 만들어진 AI 인프라 회사입니다.
- 특히 GPU 대신 자체 개발한 LPU(Language Processing Unit) 칩을 사용하여 LLM 추론 속도를 크게 높인 것이 특징입니다.

Groq 특징

항목	설명
주요 사업	AI Inference(추론) 플랫폼
자체 칩	LPU (Language Processing Unit)
특징	매우 낮은 지연시간(Low Latency), 높은 Token/sec
API 제공	OpenAI와 유사한 API 제공
지원 모델	Llama, Qwen, Gemma 등 다양한 오픈소스 모델
사용 목적	챗봇, AI Agent, 음성 AI, 실시간 서비스

Groq 모델

Groq 정책에 따라 사용가능한 Models은 변경될 수 있음

groq

Personal / Default Project

Playground API Keys Dashboard Docs

Q Search CTRL K

Docs

API Reference

GETTING STARTED

Overview

Quickstart

Models

OpenAI Compatibility

Rate Limits

Templates

API Reference

Production Models

Note:

 Production models are intended for use in your production environments. They meet or exceed our high standards for speed, quality, and reliability. Read more [here](#).

MODEL ID	SPEED (T/SEC)	PRICE PER 1M TOKENS	RATE LIMITS (DEVELOPER PLAN)	CONTEXT WINDOW (TOKENS)	MAX COMPLETION TOKENS	MAX FILE SIZE
 Llama 3.1 8B llama-3.1-8b-instant	560	\$0.05 input \$0.08 output	250K TPM 1K RPM	131,072	131,072	-
 Llama 3.3 70B llama-3.3-70b-versatile	280	\$0.59 input \$0.79 output	300K TPM 1K RPM	131,072	32,768	-
 GPT OSS 120B openai/gpt-oss-120b	500	\$0.15 input \$0.60 output	250K TPM 1K RPM	131,072	65,536	-

On this page

Featured Models and Systems

Production Models

Production Systems

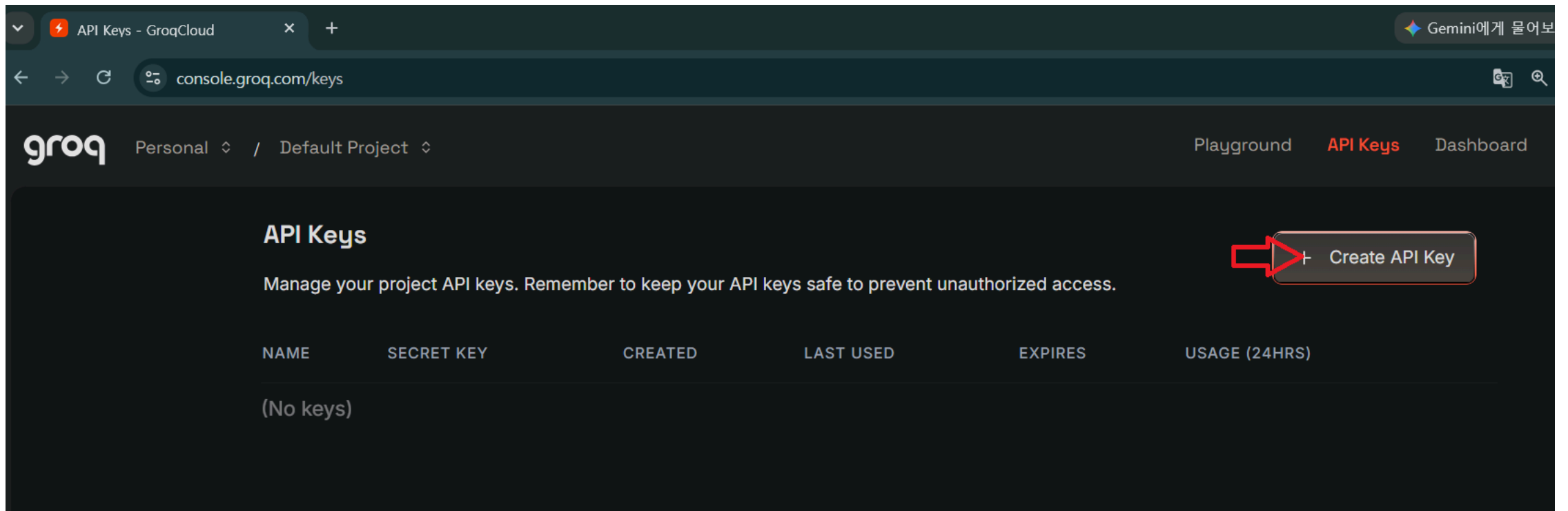
Preview Models

Deprecated Models

Get All Available Models

테스트

Groq API Key 발급



The screenshot shows the Groq console interface. The browser tab is labeled 'API Keys - GroqCloud' and the address bar shows 'console.groq.com/keys'. The Groq logo is in the top left, and navigation links for 'Personal', 'Default Project', 'Playground', 'API Keys', and 'Dashboard' are in the top right. The main heading is 'API Keys' with a subtitle: 'Manage your project API keys. Remember to keep your API keys safe to prevent unauthorized access.' A red arrow points to a button labeled '+ Create API Key'. Below this is a table with columns: NAME, SECRET KEY, CREATED, LAST USED, EXPIRES, and USAGE (24HRS). The table currently shows '(No keys)'.

NAME	SECRET KEY	CREATED	LAST USED	EXPIRES	USAGE (24HRS)
(No keys)					

Create API Key



Display Name

test

A display name for the key. Maximum 50 characters.

Expiration

No expiration

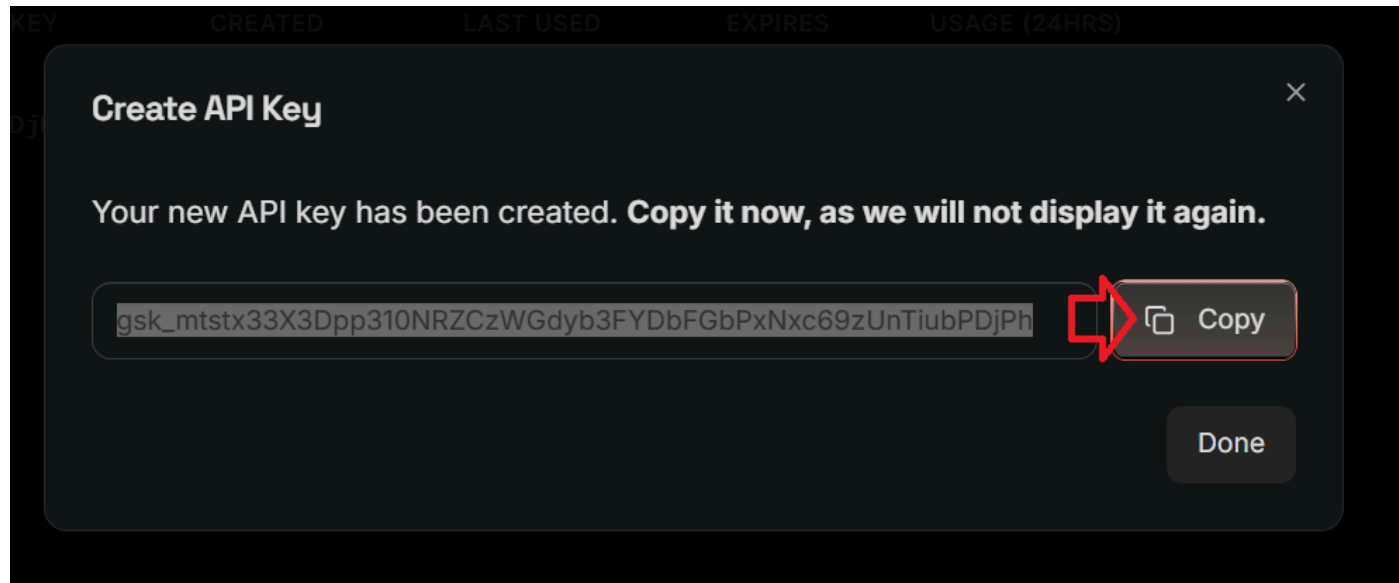


This key will not expire.

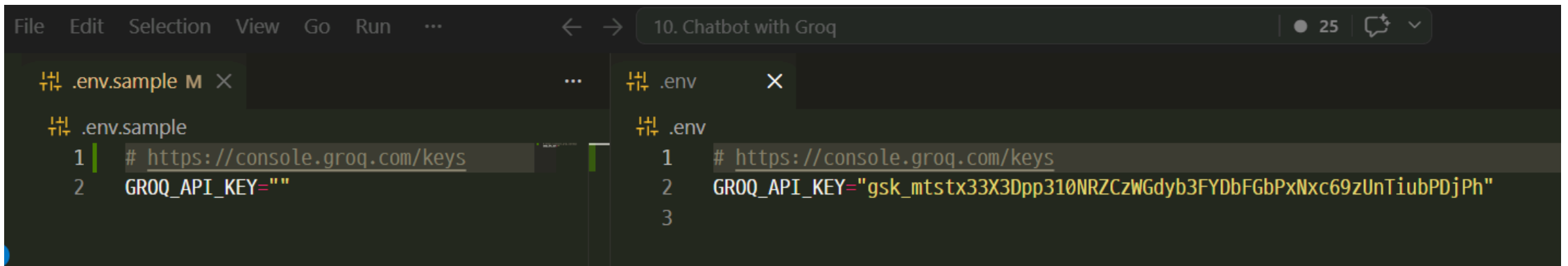


Submit

생성된 API Key 복사



.env 파일 생성 복사한 API Key 적용



가상환경

1. 챗봇 프로젝트 폴더로 이동
2. 가상환경 만들기

```
uv venv .venv --python 3.12
```

3. 가상환경 접속

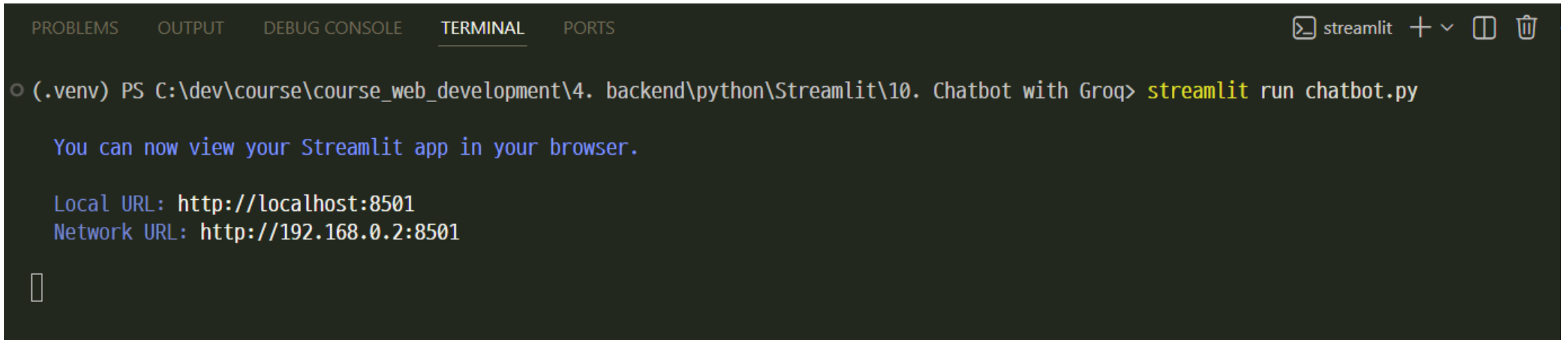
```
.\.venv\Scripts\activate
```

4. 라이브러리 설치

```
uv pip install -r .\requirements.txt
```

챗봇 실행

```
streamlit run chatbot.py
```



The screenshot shows a VS Code terminal window with the following content:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS streamlit + v [ ] [X]
```

○ (.venv) PS C:\dev\course\course_web_development\4. backend\python\Streamlit\10. Chatbot with Groq> streamlit run chatbot.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.0.2:8501

[]

실행 결과

